Esau Hervert
Math 6313

Homework 2.1

Problem 3.b

$$x = \frac{2}{7} = .\overline{285714}_{10}$$

To turn the number into binary:

$$\begin{array}{cc}
.285714285714\ldots_{10} & \\
*\,2 & \\
\hline
\mathbf{0}.571428571428\ldots_{10} & a_1 = 0 \\
*\,2 & \\
\hline
\mathbf{1}.142857142857\ldots_{10} & a_2 = 1 \\
*\,2 & \\
\hline
\mathbf{0}.285714285714\ldots_{10} & a_3 = 0 \\
*\,2 & \\
\hline
\mathbf{0}.571428571428\ldots_{10} & a_4 = 0 \\
\ldots & \ldots
\end{array}$$

$$\therefore x = 0.01001001\overline{001}_2$$

$$x = 1.001001001001001001001001001\overline{001}_2 * 2^{-2}$$

$$x_- = 1.001001001001001001001001001$$

$$x_+ = 1.001001001001001001001001010 * 2^{-2}$$

$$x - x_- = .\overline{001} * 2^{-26}$$

$$x_+ - x = 1.0 * 2^{-25} - 1.\overline{001} * 2^{-26}$$

To subtract these numbers:

$$\begin{array}{r}
0.111111\overline{111} * 2^{-25} \\
-0.100100\overline{100} * 2^{-25} \\
\hline
0.011011\overline{011} * 2^{-25}
\end{array}$$

$$\therefore x_+ - x = 0.\overline{110} * 2^{-26}$$

Thus, we can see that x is closest to x_-. The relative error is given by:

$$\frac{|x - x_m|}{|x|} = \frac{|x - x_-|}{|x|} = \frac{1.\overline{001} * 2^{-29}}{1.\overline{001} * 2^{-2}} = 2^{-27} \approx 7.45 * 10^{-9}$$

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Problem 4

Is $x = \frac{2}{3}(1 - 2^{-24})$ a machine number on the Marc-32?

$$x = \frac{2}{3} - \frac{2}{3} * 2^{-23}$$

$$\frac{2}{3} = .\overline{666}$$

$$\frac{2}{3} = 0.10101010101\overline{01}_2$$

$$\frac{2}{3} - \frac{2}{3} * 2^{-24} = 0.\overline{10}_2 - 0.\overline{10}_2 * 2^{-23}$$

$$\begin{array}{r} 0.101010101010101010101010101010\ \ldots \\ -\mathbf{0.0000000000000000000000000010}\ \ldots \\ \hline 0.1010101010101010101010101010000 \end{array}$$

From here we can now see that x is equal to:

$$1.01.01010101010101010101010101000 * 2^{-1}$$

Which is in fact a machine number on the Marc-32.

Problem 7

How many normalized machine numbers are there on the Marc-32?

Numbers on the Marc-32 have the form:

$$\pm 1.a_1 a_2 \ldots a_{23} * 2^m$$

Each a_i has the possibility of being either a 0 or a 1. Thus for $1.a_1 a_2 \ldots a_{23}$, there are 2^{23} possibilities. The value m ranges from -126 to 127 thus we have 254 different values for m making our total possibilities $254 * 2^{23}$. Since each number can be positive or negative, we have that the total normalized machine numbers is $254 * 2^{24}$.

Problem 9

Let $x = 1.111111111111111111111111111110000_2 * 2^{16}$

$$x_+ = 1.0 * 2^{17}$$

$$x_- = 1.11111111111111111111111 * 2^{16}$$

$$x_+ - x \Rightarrow$$

$$\begin{array}{r} 0.1111111111111111111111111111\ \ldots * 2^{16} \\ -0.1111111111111111111111111100\ \ldots * 2^{16} \\ \hline 0.00000000000000000000000000\overline{11} * 2^{16} \end{array}$$

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$$\Rightarrow 1.\overline{11} * 2^{-11} = 1.0 * 2^{-10}$$

$$x - x_{-} \Rightarrow 0.000000000000000000000011100 * 2^{16} = 1.1100 * 2^{-7}$$

$$\therefore fl(x) = 1.0 * 2^{27}$$

$$x_{+} - x_{-} = 1.0 * 2^{-7}$$

$$\frac{|x - fl(x)|}{|x|} = \frac{1.0 * 2^{-10}}{1.11111111111111111111111111000 * 2^{16}}$$

$$= \frac{2^{-10}}{2^{17} - 2^{-10}} = \frac{1}{2^{10}(2^{17} - 2^{-10})} = \frac{1}{2^{27} - 1} \approx 7.45 * 10^{-9}$$

Problem 14

Note:

$$fl(x \odot y) = [x \odot y](1 + \delta)$$

Which of the following is not necessarily true on the Marc-32?

a)

$$fl(xy) = [xy](1 + \delta)$$

This is true of machine operations.

b)

$$fl(x + y) = [x + y](1 + \delta)$$

True by same reason as a).

c)

$$fl(xy) = \frac{xy}{1 + \delta} = \frac{xy(1 - \delta)}{1 - \delta^2}$$

Note that δ^2 is so small that we may go ahead and assume $\delta^2 \approx 0$. Thus we can do the following:

$$\frac{xy(1 - \delta)}{1 - \delta^2} \approx xy(1 - \delta)$$

Thus, this equality is true.

d)

$$|fl(xy) - xy| \leq |xy| * 2^{-24}$$

$$fl(xy) = xy(1 + \delta) = xy + xy\delta$$

$$fl(xy) - xy = xy + xy\delta - xy = xy\delta$$

$$\therefore |fl(xy) - xy| = |xy\delta| = |xy| * |\delta| \leq |xy| * 2^{-24}$$

Thus, this inequality is true.

e)

$$fl(x+y+z) = [x+y+z](1+\delta)$$

$$fl(x+y+z) = fl\big((x+y)+z\big) = [fl(x+y)+z](1+\delta_1) = \big[[x+y](1+\delta_2)+z\big](1+\delta_1)$$

$$= [x+y+x\delta_2+y\delta_2+z](1+\delta_1) = x+y+x\delta_2+y\delta_2+z+x\delta_1+y\delta_1+x\delta_1\delta_2+y\delta_1\delta_2+z\delta_1$$

$$= [x+y+z]+\delta_1[x+y+z]+\delta_2[x+y]+\delta_1\delta_2[x+y]$$

$$= [x+y+z](1+\delta_1)+\delta_2[x+y]$$

This does not give us the inequality we were looking for; thus, the given equality is false.

Problem 16

Are these numbers on the Mark 32?

a)

$$10^{40} = 1.0 * 10^{40}$$

Given that the largest number that the Marc-32 can get is $2^{128}-1 \approx 3.4028234 * 10^{38}$, we see that 10^{40} is too large a number for the machine.

b)

$$2^{-1}+2^{-26} = 1*2^{-1}+1*2^{-26} = 2^{-1}(1+1*2^{-25}) = 1.0000000000000000000000001 * 2^{-1}$$

Since this number would require 25 places after the decimal and the Marc-32 can only go to 23, this number is not a number on this machine.

c)

$$\frac{1}{5} = .2_{10} = 0.00110011\overline{0011}$$

Since this is a repeating decimal number, the Marc-32 cannot get the exact number thus this is not a number on this machine.

d)

$$\frac{1}{3} = .\bar{3}_{10} = 0.0101\overline{01}$$

For the same reason as c), this is not a number on the Marc-32.

e)

$$\frac{1}{256} = \frac{1}{2^8} = 2^{-8} = 1.00\ldots00 * 2^{-8}$$

Since we can write this number in the form that can be stored on the Marc-32, we have that this number is in fact storable on the Marc-32.